PROJECT REPORT

Cloud-Based PayrollManagement System

Responsive role-based payroll and workforce management interface

Project Attribute	Details
Project Type	Frontend-based cloud payroll and workforce management interface
User Roles	Admin, Manager, Employee (User)
Core Technologies	HTML5, CSS3, Bootstrap 5, JavaScript, Bootstrap Icons, Google Fonts (Inter)
Submitted Implementation	19 HTML pages organized by role
Primary Focus	Payroll, tasks, attendance, leave, dashboards, approvals, profiles, and settings
Source Documents	Overview.pdf and bundled SRS.pdf

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1. Executive Summary

The Cloud-Based Payroll Management System is a responsive web interface designed to present payroll and workforce management operations through role-specific dashboards and workflows. The submitted implementation contains separate Admin, Manager, and Employee interfaces and covers payroll, task management, attendance, leave management, approvals, profiles, settings, and related dashboards.

The project is intentionally centered on the frontend layer. The bundled SRS states that backend processing, active database implementation, and live cloud hosting are outside the current development scope. The implementation therefore uses static or dummy data to demonstrate the required user journeys and business-rule visualizations.

The project requirements describe a performance-linked payroll model in which 50% of salary is released by default and the remaining 50% is associated with approved task completion. Approved completion of 70% or more releases 100% of salary; 50%-69% temporarily retains two days of salary; and below 50% retains seven days of salary until compliance or managerial decision. These rules are incorporated into the report as the central business logic the UI is designed to represent.

2. Introduction

2.1 Background

Traditional payroll workflows often require coordination across attendance, task completion, leave records, approvals, and salary processing. The project addresses this coordination problem by presenting a unified payroll and workforce management interface with role-based access and workflow-oriented screens.

2.2 Purpose

The purpose of the system is to provide an intuitive, responsive interface for visualizing and managing salary disbursement, attendance validation, task-based compensation, leave management, reporting, and approval workflows.

2.3 Intended Audience

Project developers and frontend engineers

UI/UX designers

Quality assurance and testers

Administrators and managers

Employees / end users

3. Objectives

Provide a modern and responsive payroll-management user interface for desktop and mobile-sized screens.

Separate system functionality according to Admin, Manager, and Employee responsibilities.

Represent task-based salary control and salary retention rules through clear payroll views and approval workflows.

Provide attendance workflows that represent remote check-ins, geo-location, timestamps, device data, and Manager approval status.

Provide leave workflows including monthly paid leave credit, rollover, annual cutoff/reset, and approval records.

Provide dashboards, tables, cards, forms, filters, and navigation patterns that support day-to-day workforce administration.

Create a frontend foundation that can later be connected to backend services, a database, authentication, cloud storage, and production security controls.

4. Scope and Project Boundaries

4.1 Included in the Current Implementation

Role-based HTML pages for Admin, Manager, and Employee

Responsive layouts and dashboard interfaces

Payroll and salary views

Task assignment and task progress views

Attendance and approval screens

Leave management and review screens

Employee and manager management interfaces

Profile and system settings interfaces

Interactive client-side UI elements using JavaScript and Bootstrap components

4.2 Explicitly Outside the Current Scope

Backend processing and server-side business logic

Live database integration

Production authentication/authorization infrastructure

Live cloud deployment

Real payroll disbursement or banking integration

Production GPS verification

Production automated backup execution.

5. System Overview

The system is organized around three roles. Each role receives a dedicated navigation structure and role-specific screens so that users see only the workflows relevant to their responsibilities.

Role	Primary Responsibilities	Implemented UI Areas
Admin	Full system control; employee onboarding/offboarding; payroll configuration; reporting and audit views	Dashboard, Employees, Managers, Payroll, Attendance, Tasks, Settings, Profile
Manager	Task assignment and progress; attendance validation; leave review; salary approval	Dashboard, Assign Task, Approve Attendance, Review Leave, Salary Approval
Employee	Attendance; salary visibility; leave usage; task progress; profile access	Dashboard, Attendance, Salary, Leave, Tasks, Profile

6. Technology Stack

Technology	Role in the Project
HTML5	Structure of dashboards, forms, tables, navigation, cards, and module pages
CSS3	Custom visual styling, spacing, responsive behavior, page-specific UI treatments, and design consistency
Bootstrap 5	Responsive grid, components, utilities, tables, forms, navigation and layout support
JavaScript	Client-side interactions, UI behavior, data presentation logic, filters, modals and dynamic states
Bootstrap Icons 1.11.1	Interface icons used across navigation and actions
Google Fonts - Inter	Typography used for the interface design

7. User Roles and Access Hierarchy

7.1 Admin

The Admin role represents the highest level of system control. The UI supports employee and manager management, payroll configuration, task management, attendance visibility, settings, profile management, and dashboard-level oversight.

7.2 Manager

The Manager role is centered on team operations and approval workflows. It includes task assignment, task progress handling, attendance approval, leave review, and salary approval. The Overview requires only Managers to approve completed tasks and to approve attendance and leave usage.

7.3 Employee

The Employee role provides self-service access to attendance, salary, leave, tasks, dashboard information, and profile data. The employee can initiate attendance and view their current payroll-related information, while approval actions remain with the Manager where specified.

8. Functional Modules

Module	Purpose	Implemented Pages
Dashboard Module	Role-based summary information including workload, attendance, leave, salary, and performance indicators.	Admin/Manager/Employee dashboards
Employee Management	Employee listing, status/summary information, profile-related views, and onboarding/offboarding support at the UI level.	Admin Employees
Manager Management	Manager records and role-related workforce information.	Admin Managers
Payroll & Salary	Salary summaries, payroll cycle information, release percentages, deductions/retentions, and salary approval workflow.	Admin Payroll, Manager Salary Approval, Employee Salary
Task Management	Task listing, assignment, progress representation, and approval-related workflow.	Admin Tasks, Manager Assign Task, Employee Tasks
Attendance	Attendance records, remote check-in representation, pending approval states, and attendance oversight.	Admin Attendance, Manager Approve Attendance, Employee Attendance
Leave Management	Leave ledger, leave balance, leave usage workflow, and manager review.	Manager Review Leave, Employee Leave
System Settings	Company information, payroll settings, leave settings, and attendance settings.	Admin Settings
Profile	Role-specific user profile information.	Admin Profile, Employee Profile

9. UI/UX Design and Eight Major Interface Styles

The project contains multiple page treatments and layout patterns rather than one repeated screen. Based on the submitted implementation and the project description, the interface can be grouped into approximately eight major UI styles:

UI Style	Characteristics
1. Dashboard Style	KPI cards, welcome/header area, charts or summaries, status indicators, and quick operational insights.
2. Management/Table Style	Data tables, filters, badges, row actions, employee/manager records, and summary counters.
3. Payroll/Financial Style	Salary figures, payroll cycle summaries, release percentages, retention indicators, and approval-oriented presentation.
4. Task Workflow Style	Task lists, assignment forms, progress percentages, priority/status indicators, and approval states.
5. Attendance/Location Style	Attendance records, date/time information, pending approval states, and location/check-in representation.
6. Leave/Ledger Style	Leave balance, monthly credit, ledger/history, application state, and approval/review workflow.
7. Profile/Settings Style	Profile information, company settings, payroll configuration, leave settings, and attendance configuration forms.
8. Approval/Review Style	Manager decision screens for attendance, leave, and salary approvals with explicit status and action controls.

Across these styles, the interface emphasizes responsive layouts, sidebar navigation, dashboard cards, tables, forms, and modern web-browser compatibility, matching the SRS interface requirements.

10. Business Rules and Payroll Control Logic

The Overview defines a performance-linked payroll mechanism. The frontend is expected to visualize these rules clearly; actual salary processing would require backend enforcement in a production system.

10.1 Base Salary Release Rule

Every employee receives 50% of salary by default. The remaining 50% is linked to approved task completion.

10.2 Performance Thresholds

Approved Work Completion	Payroll Outcome	Interpretation
70% or more	100% salary released	Full salary release when sufficient approved work is completed
50% to 69%	2 days salary retained temporarily	Retained amount is released with the next cycle after completion
Below 50%	7 days salary retained	Retention continues until compliance or a managerial decision

10.3 Task Workflow Rules

Tasks may be assigned by Admin, Manager, or User according to the project requirements.

Only the Manager can approve completed tasks.

Task activities should be recorded with timestamps.

Task details are limited to a title and textual description; no subtask hierarchy is required.

Task progress is represented using percentages such as 10%, 20%, and onward.

A production payroll engine should implement this logic server-side so that salary release and retention cannot be altered by client-side manipulation. In the submitted project, these rules are represented at the interface level.

11. Attendance and Leave Policies

11.1 Attendance Policy

Employees can mark attendance remotely from any location.

Precise geographic coordinates are to be recorded.

Geo-location, timestamp, and device data are stored as part of the attendance record.

Attendance remains pending until Manager approval.

The workflow supports accountability for field staff and remote teams.

11.2 Leave Management Policy

One paid leave is credited each month.

Unused paid leave rolls over monthly.

Annual leave balance resets/cuts off on 1 April.

Manager approval is required for leave usage.

A complete leave ledger is maintained.

12. Security, Cloud, Backup, and Audit Requirements

Requirement from Overview	How It Applies to the Project	Current Status
Cloud-hosted web platform	Target deployment model for the payroll platform	Not implemented in submitted frontend
SSL-secured access	HTTPS should protect production access	Production infrastructure not included
Daily automated backups	Operational backup strategy for payroll/workforce records	Backend/storage not included
Role-based permissions	Admin, Manager, Employee responsibilities are separated in the UI	Represented in role-based pages; backend enforcement pending
Audit trails	Payroll and approval activity should be timestamped and traceable	UI/workflow concept represented; persistent audit storage pending
Mobile-friendly responsive interface	All role pages should function across desktop and mobile screen sizes	Included as a frontend design goal

The distinction between UI representation and production security is important. Role-based navigation in HTML demonstrates the intended access hierarchy but does not constitute secure authorization. A deployed system must enforce authorization, encryption, audit persistence, and backup controls on the server side.

13. System Architecture and Data Representation

13.1 Current Frontend Architecture

Browser-based HTML pages grouped by user role

Bootstrap-based responsive layout layer

Custom CSS for page-specific and shared styling

JavaScript for client-side interactions and dynamic UI states

Static or dummy data for demonstrations

13.2 Logical Data Model Represented by the SRS

Entity	Key Fields / Information
Employee	Employee ID, Name, Role (Admin/Manager/User)
Task	Task ID, Title, Textual Description, Progress Percentage
Attendance	Date, Geo-location, Timestamp, Device Data, Status (Pending/Approved)

13.3 Production Architecture Recommendation

For production use, the frontend should connect to authenticated backend APIs, a relational database, secure cloud storage where required, a payroll rules engine, an audit-log service, and scheduled backup infrastructure. These components are future integration targets rather than part of the submitted frontend implementation.

14. Requirements Traceability

Requirement	Evidence in Project	Assessment
Three-role access hierarchy	Admin / Manager / Employee pages	Implemented at UI level
Dashboard module	Role-specific dashboard pages	Implemented
Payroll engine visualization	Admin payroll + Manager salary approval + Employee salary	Implemented at UI level
Task assignment and approval	Manager assign-task + task pages	Implemented at UI level; approval enforcement requires backend
Task progress in percentages	Task interfaces and dashboard indicators	Represented in UI
Remote attendance	Employee attendance + Manager approval + Admin attendance	Represented in UI
Geo-location / timestamp / device	Attendance requirements in SRS/Overview	Target data model; live capture/storage not implemented
Leave ledger and approval	Employee leave + Manager review leave	Implemented at UI level
1 paid leave per month	Leave policy	Target/business rule represented by requirements; backend calculation pending
1 April reset/cutoff	Leave policy	Target/business rule
Salary retention thresholds	Payroll / salary views	Target/business rule represented; backend enforcement pending
Role-based security	Separate role UIs	UI representation only
Daily backups / SSL / cloud hosting	Overview infrastructure requirements	Not implemented in frontend package
Responsive mobile-friendly interface	Bootstrap/CSS responsive design	Implemented as frontend design requirement

15. Testing and Validation Approach

Because the submitted project is frontend-focused, validation should concentrate on navigation, visual consistency, responsive behavior, form interactions, role separation, and correctness of displayed business rules.

Test Area	Example Validation	Expected Result
Navigation	Open index.html and move through each role flow	Correct pages open and role-specific navigation is coherent
Role UI separation	Review Admin, Manager and Employee menus	Each role exposes appropriate workflows
Payroll rule display	Check 70%+, 50%-69%, and <50% cases in payroll-related views	Correct outcome text/status is presented
Task workflow	Assign task, update progress, review task state	Progress and status indicators update as designed
Attendance workflow	Review pending/approved states and remote check-in representation	Attendance states are visible and manager approval is clear
Leave workflow	Review balance, ledger, application/review states	Policy information and approval workflow are understandable
Responsive behavior	Resize to mobile and tablet dimensions	Layout remains usable without horizontal overflow
UI consistency	Compare cards, tables, buttons, icons, spacing, and navigation	Design language remains consistent across page styles
Accessibility baseline	Keyboard navigation, contrast, labels, headings	Core controls remain understandable and operable

16. Limitations

The submitted implementation is frontend-only and does not provide production backend processing.

No live database integration is included; data is static or dummy data for demonstration.

The project package does not provide live cloud deployment infrastructure.

Actual payroll disbursement, retention enforcement, bank integration, and accounting reconciliation are not performed by the current interface.

GPS and device information are represented as interface/data concepts rather than verified production telemetry.

SSL, automated backups, and persistent audit logging require server/cloud infrastructure outside the current frontend scope.

Client-side role navigation should not be treated as secure authorization without backend permission checks.

17. Future Enhancement Roadmap

Build REST APIs and connect the frontend to a secure backend service.

Add a relational database for employees, managers, payroll cycles, tasks, attendance, leaves, approvals, and audit logs.

Implement secure authentication with server-side role and permission enforcement.

Move payroll calculation and retention rules into a server-side payroll engine.

Integrate verified GPS/location capture with privacy and consent controls.

Introduce automated cloud backups, monitoring, SSL certificates, secrets management, and recovery procedures.

Add exportable payroll reports, audit reports, and analytics dashboards.

Introduce automated unit, integration, UI, and security testing for the production system.

18. Conclusion

The Cloud-Based Payroll Management System provides a structured frontend foundation for a modern payroll and workforce management platform. It separates responsibilities across Admin, Manager, and Employee roles and presents dedicated interfaces for payroll, tasks, attendance, leave, profiles, approvals, and system settings.

The most important business requirement captured by the project is performance-linked salary control: 50% is released by default, while the remaining 50% is linked to approved task completion, with defined retention rules based on completion thresholds. The attendance and leave policies further connect workforce activity to payroll transparency and managerial oversight.

As delivered, the project should be considered a frontend implementation/prototype aligned to the stated cloud payroll requirements. Its next major step is backend and cloud integration so that the UI rules can become enforceable, auditable, secure, and operational in a real organization.

Appendix A. Implemented Page Inventory

Role	Page	Purpose / UI Area
Admin	dashboard.html	Admin dashboard
Admin	attendance.html	Attendance & leave management
Admin	employees.html	Employee management
Admin	managers.html	Manager management
Admin	payroll.html	Payroll management
Admin	profile.html	Admin profile
Admin	settings.html	System settings
Admin	tasks.html	Task management
Manager	dashboard.html	Manager dashboard
Manager	approve-attendance.html	Attendance approval
Manager	assign-task.html	Task assignment
Manager	review-leave.html	Leave review / ledger
Manager	salary-approval.html	Salary approval
Employee	dashboard.html	Employee dashboard
Employee	attendance.html	Employee attendance
Employee	leave.html	Leave management

Employee	profile.html	Employee profile
Employee	salary.html	Salary and payroll
Employee	tasks.html	Employee tasks
Shared	index.html	Entry/login interface / role access entry point